

Aria Smart Thermostat

Software Requirements Specification (fabricated example document)

3.1 Power Management

REQ-DEMO-001 Low-power standby

Description: The thermostat shall enter standby mode after 120 seconds of no user interaction and no active heating or cooling.

Rationale: Reduces average daily power draw for battery-backed installations.

Priority: High

REQ-DEMO-002 Battery backup runtime

Description: On mains power loss, the thermostat shall maintain display and sensor operation for at least 4 hours from the time of power loss.

Rationale: Prevents loss of schedule and setpoint state during short outages.

Priority: Medium

REQ-DEMO-003 Charge indicator

Description: The thermostat shall display a battery charge indicator whenever charge is below 20%.

Rationale: Gives the user advance warning before backup power is exhausted.

Priority: Low

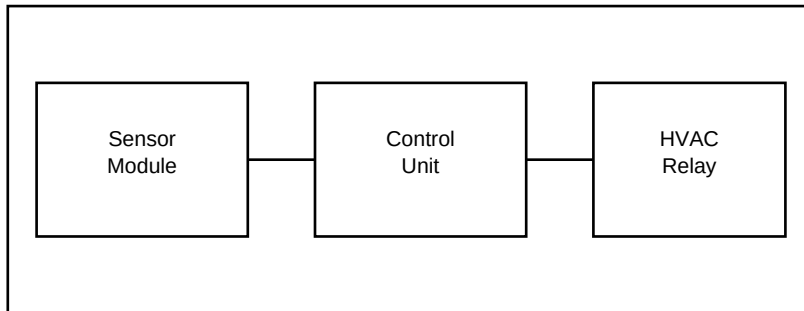


Figure 1: System Overview

3.2 User Interface

REQ-DEMO-004 Setpoint adjustment

Description: The user shall be able to adjust the target temperature in 0.5 degree increments using the front-panel dial.

Rationale: Matches the granularity users expect from comparable thermostats.

Priority: High

REQ-DEMO-005 Display readability

Description: The display shall remain readable under direct sunlight of up to 10,000 lux.

Rationale: Thermostats are frequently installed near windows or glass doors.

Priority: Medium

REQ-DEMO-006 Child lock

Description: The thermostat shall support a child-lock mode that disables the front-panel controls until a 3-second button-press.

Rationale: Prevents accidental setpoint changes in households with young children.

Priority: Low

3.3 Connectivity

REQ-DEMO-007 Wi-Fi provisioning

Description: The thermostat shall support Wi-Fi provisioning via a companion mobile app using WPA2 or WPA3.

Rationale: Required for remote schedule management and firmware updates.

Priority: High

REQ-DEMO-008 Offline schedule fallback

Description: If connectivity is lost, the thermostat shall continue to run the last synchronized schedule without interruption.

Rationale: Loss of Wi-Fi must never affect core climate-control behavior.

Priority: High

3.4 Diagnostics

REQ-DEMO-009 Sensor self-test

Description: On power-up, the thermostat shall run a self-test of the temperature and humidity sensors and report a fault.

Rationale: Silent sensor failure would otherwise show as a plausible but wrong reading.

Priority: High

REQ-DEMO-010 Diagnostic log retention

Description: The thermostat shall retain the last 30 days of fault codes and HVAC-relay activations in local non-volatile storage.

Rationale: Lets a technician review recent history during a service visit without needing cloud connectivity.

Priority: Medium

REQ-DEMO-011 Relay cycle counter

Description: The thermostat shall count and persist the total number of HVAC-relay activation cycles over the device's lifetime.

Rationale: Relay wear is a known field-failure mode; a cycle count supports predictive maintenance.

Priority: Low

3.5 Compliance

REQ-DEMO-012 Standby power limit

Description: The thermostat shall draw no more than 1.0 W average power in standby mode.

Rationale: Required to meet the target energy-efficiency labeling class.

Priority: High

REQ-DEMO-013 Operating temperature range

Description: The thermostat shall operate correctly across an ambient range of 0C to 40C.

Rationale: Covers unconditioned hallway and utility-room installations.

Priority: Medium

REQ-DEMO-014 RF emissions

Description: The thermostat's Wi-Fi radio shall comply with the applicable regional RF emissions and coexistence standards.

Rationale: Required for regulatory certification prior to sale.

Priority: High